Quantum Hardware-Embedded Newton-Raphson Method for Power Flow Calculation

APPENDIX
DATA OF 6-BUS ISLANDED MICROGRID SYSTEM

The topology of 6-bus islanded microgrid system is shown in Fig. 1 and the branch parameters are listed in Table I. In the table, R is the resistance of the power line, X is the reactance of the power line, and B is the total line charging susceptance.

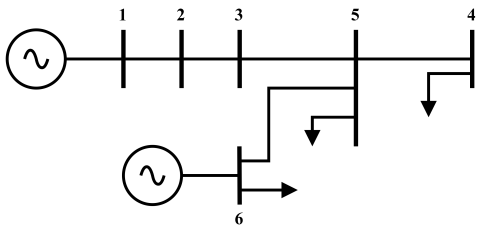


Fig. 1. The 6-bus islanded microgrid system.

TABLE I
BRANCH PARAMETERS OF THE 6-BUS ISLANDED MICROGRID SYSTEM

From Bus	To Bus	R	X	B
1	2	0.0000	0.0576	0.0000
2	3	0.0170	0.0920	0.1580
3	4	0.0390	0.1700	0.3580
3	5	0.0119	0.1008	0.2090
3	6	0.0085	0.0072	0.1490

Buses 1 and 6 are the droop-controlled buses, whose model is given by [1]:

$$\begin{cases} P_{G,i} = \dfrac{1}{m_{p,i}} \left(\omega_0 - \omega\right) \\ Q_{G,i} = \dfrac{1}{n_{q,i}} \left(U_{0,i} - U_i\right) \end{cases} \tag{1}$$

where $P_{G,i}$ and $Q_{G,i}$ are the active and reactive power output of the generator at bus i; $m_{p,i}$ and $n_{q,i}$ are the droop coefficients; ω_0 and $U_{0,i}$ are the frequency reference and nominal voltage magnitude of bus i; ω is the frequency of the microgrid; U_i is the voltage magnitude of bus i.

Moreover, the static load at bus i is modeled as:

$$\begin{cases} P_{L,i} = P_{0,i} U_i^{\alpha} \left(1 + k_{p,i} \left(\omega - \omega_1\right)\right) \\ Q_{L,i} = Q_{0,i} U_i^{\beta} \left(1 + k_{q,i} \left(\omega - \omega_1\right)\right) \end{cases} \tag{2}$$

where ω_1 is the nominal frequency for loads; $P_{L,i}$ and $Q_{L,i}$ are the active and reactive power consumed by the load at bus i; $P_{0,i}$ and $Q_{0,i}$ are the nominal active and reactive load power; α and β are the voltage exponents of the loads; $k_{p,i}$ and $k_{q,i}$ are the frequency sensitivity coefficients of the active and reactive load power.

In the case study, the parameters of the droop-controlled buses are listed in Table II. In addition, the other parameters

TABLE II
PARAMETERS OF THE DROOP-CONTROLLED BUSES

Bus	$m_{p,i}$	$n_{q,i}$	$U_{0,i}$(p.u.)	$k_{p,i}$	$k_{q,i}$
1	0.0040	0.0080	1.06	1.00	-1.00
6	0.0040	0.0800	1.06	1.00	-1.00

are given as $\omega_0 = 1.00$p.u., $\alpha = \beta = 1$, $P_{0,4} = Q_{0,4} = 0.10$p.u., $P_{0,5} = Q_{0,5} = 0.07$p.u., $P_{0,6} = 0.25$p.u., and $Q_{0,6} = 0.07$p.u.. For the ill-conditioned case, the power flow calculation of the 6-bus islanded microgrid is initialized by setting the voltage magnitude to 1.00 p.u. and the voltage angle to 0.072πrad for buses 2, 3, 4 and 5.

REFERENCES

[1] H. Wang, Z. Yan, X. Xu, and K. He, "Evaluating influence of variable renewable energy generation on islanded microgrid power flow," *IEEE Access*, vol. 6, pp. 71 339–71 349, 2018.